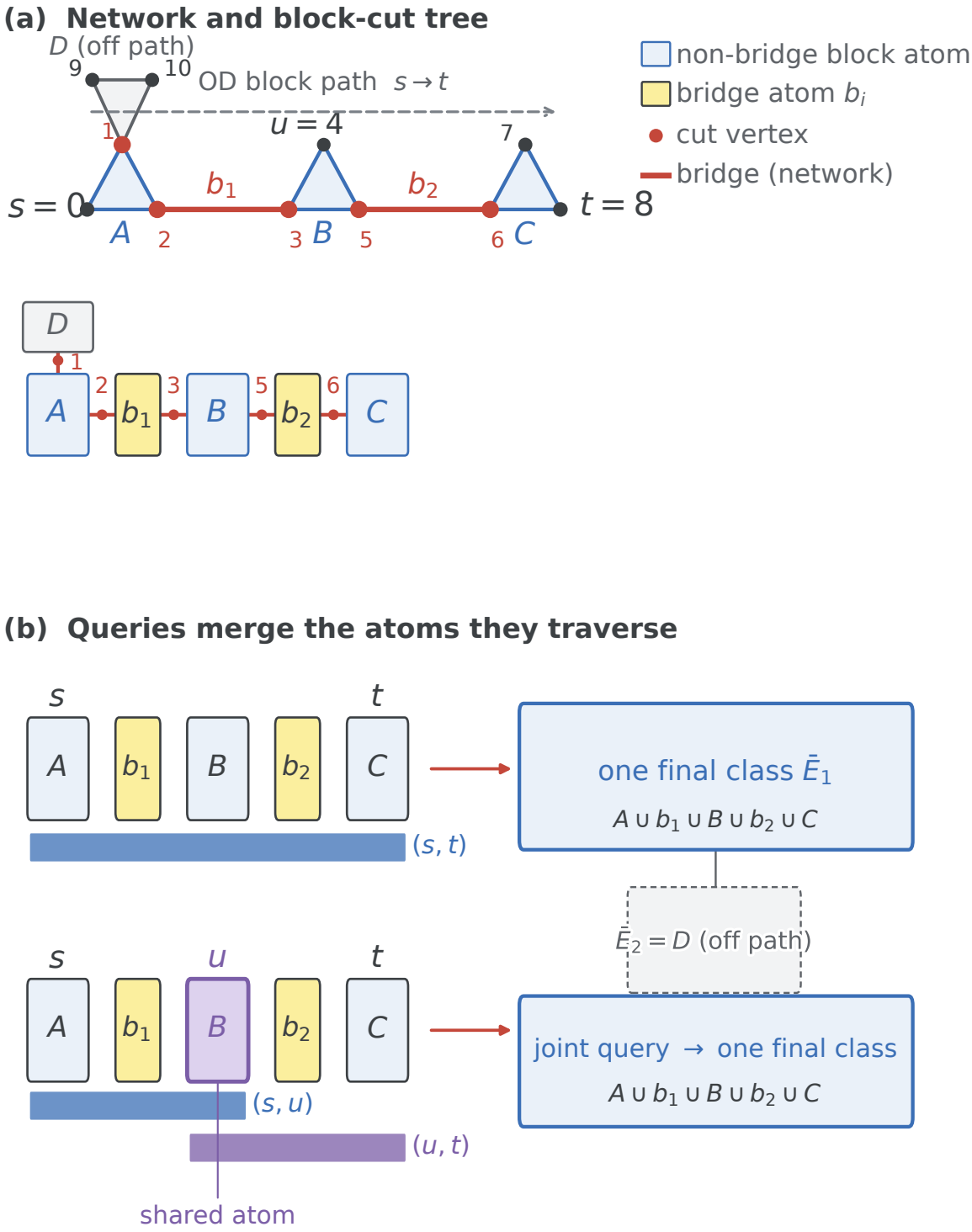
(a) Network and block-cut tree
D (off path)
9
10
OD block path s → t
1
u = 4
7
s = 0
b1
b2
t = 8
A
2
3
B
5
6
C
non-bridge block atom
bridge atom bi
cut vertex
bridge (network)
D
1
A
2
b1
3
B
5
b2
6
C
(b) Queries merge the atoms they traverse
s
t
A
b1
B
b2
C
(s, t)
one final class Ē1
A ∪ b1 ∪ B ∪ b2 ∪ C
Ē2 = D (off path)
s
u
t
A
b1
B
b2
C
(s, u)
(u, t)
shared atom
joint query → one final class
A ∪ b1 ∪ B ∪ b2 ∪ C